Mateusz Płoszaj-Mazurek

Computational designer · Grasshopper, simulation and BIM tooling · Architect, PhD

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Architect who has taught Grasshopper for seven years and builds the tools around it: plugins, simulation workflows, IFC pipelines and web viewers, for employers and clients in Denmark, Liechtenstein, Finland and Poland. I take a repetitive design task, automate it end to end and hand the team something they can run without me.

Commercial and tool work

2025 — **Free Drawing, Carbon Designer 3D** (One Click LCA)
Led a feature letting architects sketch building massing directly in the browser, from user problem through specification to release.

2024 — **IFC viewer for historical building objects** (client, Liechtenstein)
Browser viewer built with Three.js, React and TypeScript for exploring IFC models of historical building elements.

2025–26 — **OmegaFlux**
Per-point daylight prediction from geometric view descriptors with gradient-boosted trees, exported to ONNX and run in the browser. Dataset generation, held-out validation against Radiance, transfer to unseen room shapes. curiosit.github.io/OmegaFlux

2026 — **Solaie: sunlight-access checker for Revit** (own product, in testing)
Revit 2027 add-in (C#, .NET, WPF) with a compiled Python analysis engine: model scanning, geometry export, vectorised ray tracing against context meshes, signed time-limited licences and an HTML report for building authorities. solaie.pl

before 2024 — **Freelance AEC development** (clients via Upwork)
Short commercial jobs including IFC-to-web implementation.

2016–2024 — **In-house design tools** (Bjerg Arkitektur)
Grasshopper and scripting tools used in live projects, built end to end; daylight simulation studies for VELUX-related projects.

2023–24 — **goeko.gha: Ökobaudat in Grasshopper** (open source)
Grasshopper plugin that reads environmental product declarations from the German Ökobaudat database through its API. MIT licence, published on Food4Rhino. github.com/Curiosit/goeko.gha

2024 — **Wall carbon footprint configurator** (ShapeDiver)
No-code web app built from a Grasshopper definition, loading material data from Ökobaudat through an API.

2023–24 — **AEC Hackathons**
Copenhagen 2023 (SpeckleLCA), Zurich 2024 (InfraWeb), Copenhagen 2024 (OpenDetail, Best Mashup Project).

Experience

2017–now — **Researcher and lecturer**
Faculty of Architecture, Warsaw University of Technology · doctoral candidate from 2017, academic staff since 2021

- Daylight regulation, simulation and surrogate-model research; see publications.
- Teaches Grasshopper and multi-criteria building optimization (400+ students so far); runs a second-year design studio.

2024–2026 — **Software engineer, interim product owner**
One Click LCA, Helsinki · remote

- Built features for Carbon Designer 3D, an early-design tool for architects (TypeScript, React, Three.js, AG Grid).
- Interim product owner in the second half of 2025; took the Free Drawing feature from specification to release.
- Wrote methodology specifications.

2024 — **Contract developer**
enxess, Liechtenstein
Web viewer for IFC models of historical building objects (Three.js, React, TypeScript).

2023–2024

Founder

SLAD.AI

Self-funded venture building early-design carbon estimation software; designed, built and tested with users.

2014–2024

Architect, Certified Passive House Designer

Bjerg Arkitektur · Hjørring 2014–16, Warsaw 2016–24

- Single- and multi-family housing, schools, kindergartens and offices, from 100 m^2 to 15,000 m^2.
- All design phases: concept, permit and construction documentation, and architectural supervision on site.
- Code compliance in practice under Danish and Polish regulations and Passive House certification; daylight simulation studies with VELUX.
- 2022–2024: architecture, sustainability and software consultant, building in-house design tools; co-author of the Polish whole-life carbon roadmap.

Skills

Design tools	Rhino and Grasshopper, Rhino.Inside.Revit, Revit, ArchiCAD, Ladybug and Honeybee, Radiance, THERM, IFC, That Open Company tools
Methods	Evolutionary and multi-objective optimization, surrogate modelling, gradient-boosted trees, dataset and validation design, parametric experiments
Building domain	Climate-based daylight (sDA, ASE, UDI, EN 17037), insolation, Polish and Danish building regulations, Passive House, building carbon footprint
Code	Python (NumPy, pandas, scikit-learn, XGBoost, ONNX), C#, TypeScript, React, Three.js, Git, CI/CD; daily AI-assisted development with Claude Code, including six months on a production codebase
Languages	Polish (native), English (fluent), Danish (communicative), Chinese (basic)

Teaching

7 years

Introduction to Grasshopper

"Cyfrowe techniki geometryczne": parametric modelling and multi-criteria optimization of energy, daylight and carbon.

2020–

iBIM postgraduate programme (Interdisciplinary BIM Studies)

Introductory and advanced Grasshopper, environmental and daylight analysis with Rhino.Inside.Revit and Grasshopper, Python and C# scripting, for practising architects and engineers. Hybrid.

Grants and awards

2024 — **First prize, Best Mashup Project** · AEC Hackathon, Copenhagen

2022 — **PLGBC Green Building Awards: best green building without certification** · Passive House kindergarten in Podkowa Leśna, Bjerg Arkitektur; award received on behalf of the practice. Also a distinction for an educational building.

Education

2022

PhD in Architecture

Warsaw University of Technology

2014–2016

MSc in Architecture

Aalborg University, Denmark

2023–2024

Certifications

Master BIM Software Developer (That Open Company, 2024) · Harvard CS50x (2023) · Certified Passive House Designer

Talks and panels

2023 — **Pro-environmental design: training sessions for architects** · Saint-Gobain · two sessions, Jura Krakowsko-Częstochowska and Żnin, October 2023

Publications

A workflow for LLM-assisted IFC data enrichment to support circular reuse of large-panel building systems in Poland

with A. M. Tofiluk · Journal of Building Engineering 120, 115397 (2026) · doi · open access